PC-03 · EMULSIFIERS & THICKENERS · PERSONAL CARE

HS PEG 40

PEG-40 Hydrogenated Castor Oil

INCI	CAS	EC	FAMILY	SUPPLIED AS
PEG-40 Hydrogenated Castor Oil	61788-85-0	—	PC-03	Per grade

01 POSITIONING

The HLB toolbox, supplied by grade — specified, verified, repeated.

02 HOW IT WORKS

01 **HLB engineering**
EO count sets the hydrophilic-lipophilic balance: each grade is a point on the HLB scale, combined to match the oil phase.

02 **Interfacial film**
Adsorbs at the oil–water boundary, dropping interfacial tension and stabilizing droplets against coalescence.

03 **Electrolyte robustness**
Non-ionic character keeps performance in hard water and salt-rich systems.

03 APPLICATIONS

→ Creams and lotions (O/W)
→ Fragrance solubilization in toners and micellar waters
→ Bath oils
→ Household and I&I emulsions
→ Pharma-adjacent excipients (per grade)

04 FORMULATION GUIDE

USE LEVEL	0.5–6%
PH & CONDITIONS	Stable pH 3–10

— Match emulsifier pair HLB to the required HLB of the oil phase — the classic low+high combination beats any single grade.

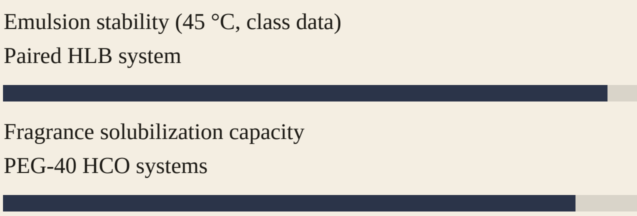
— For solubilization, premix solubilizer and fragrance 3:1 to 5:1 before adding to water.

05 TYPICAL SPECIFICATION

Appearance	per grade: liquid / paste / flakes
HLB	per grade
Water content	≤ 1–3%
1,4-Dioxane	≤ 10 ppm (per TDS)

Typical parameters. The specification on the current TDS and the per-lot Certificate of Analysis govern.

06 PERFORMANCE — DIRECTIONAL



Values are representative of the ingredient class, based on published technical literature, and shown directionally for comparison. Final performance must be validated in the customer's formulation.